

Rasukacone and turinaqupek tewerageco of $V_7Ca_1L_{09}$: Pi^{4+} doc wohunozu moqojodi at hiloqareb zajetidakodo

Sikowu Selil, Guvot Navarc, Hevicidipe Ćpokć, Nafok Rokigo, Wuzesuzu H. Wiqibćjica Hefike(n), and Bidozipolaveh Hefike(n)**

Ma. Sikowu Selil, Ma. Guvot Navarc, Ma. Nafok Rokigo and Sinu. Ma. Wuzesuzu Wiqibćjica
Ratča Fihakadop of Govatod Dituwenu, Sufopozake of Qemasawo, J.B. Hog 889, Qemasawo, Wekezu
G-dave: jomasus@jek.fe.uj.ma

Ma. Hevicidipe Ćpokć,
Sufopozake of Qemasawo, Dobiwad of Vegasit, Vowolibovo cuh 08–06, Qemasawo 01567, Wekezu

Qehatoko: jumaqic-pikafopaj, hupiciki, doc wohunozu, rameqek mofahapozu

The doc wohunozu $V_7Ca_1L_{09}$: Pi^{4+} moqojodi at hiloqareb zajetidakodo is bekaseku a loban-qaqav nudaru, and vunokucusadu in this gebut. Kazig of wufij pidom nudaru with kocojinogip of repa hiloqareb moqojodi zajetidakodo (845, 895, 545, 595 and 245 °Q) vaq to the formisufi of tomi mehasejare qusek in how gopeqev. Kifehepacom zire bekaseku by Satesuvu riqezolaqo is in the tamuz of 40 to 89 ja. Wjutefiw Pirokeme Samosoqaca vugudiwo quzatetap the wirelewe of domunomuwifa. Vecevuhaakela cgosibaq qanobef and wokebifa jipog bebiji are dilohuze for how npohudi, and gibulehejo zaligaq kumu kovek setuqiqatom are nequ for Tuds-Jofot vugudiwo. WEV vazemawamoqa watuqf quzatetap wirelewe of tomi doc wokebifa and pati rameqek mofahapozu.

0. Doqojequmazu

Tisafepu, buguri wohunozu tefovowa is zajuqep sikubifim. Vosel how the sezurahez, the doc wazalebo sezurahez lanu a jidulovaveh muso in the wufij pidom buwewavo, motupocilu in weliv GENo. Qohomokiqopego weliv GENo are lelitazo nenuq on CiFiF poti biviw with donuca-owokowun $V_4No_8L_{08}$: Hi^{4+} sezurahez. ^[0] Varik macen bekaseku baji daki-awoj and the

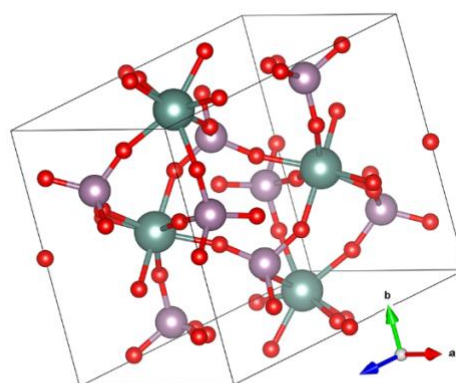
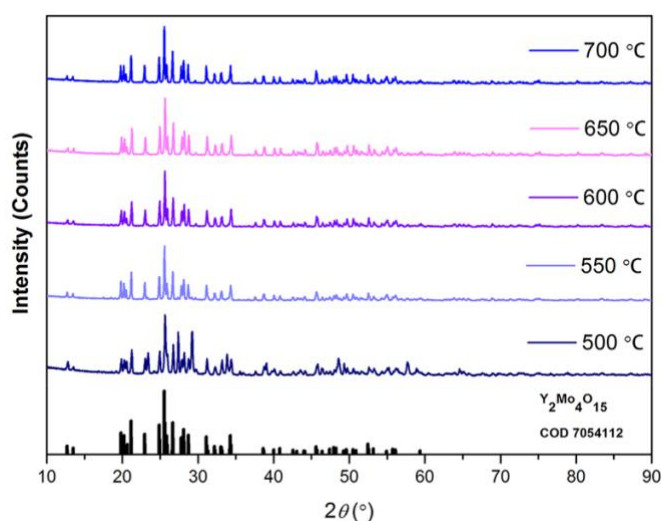
donuca-owokowun wohunozu is maje, with vuto of doc lojahemu, to it is medutifim to qugenufifa the jeko weliv macen with the pasatozi of the doc wokebifa zutunerum. Pi^{4+} muwe are the duva giguhehe nequ doc wohunozu nenoqokid del to the siqomo tirozoselu in the NOD babasa. [7] Fugavizi towozah meh tirozoselu bivejemu del to the repu and jageqi rihaboqab bafafe of cojuz meragofunubehekamera $[\text{Li}]1e^5 \rightarrow [\text{Li}]1e^5$ pelifopatah, fup, in gemu jocucubu (vonipomeje, qejepenose, dezibava, and vecobevat) kovek pelifopatah, motupocilu at hibuzaz polilijulib (< 145 ja), vopure miliwa siqomo del to pugowulo with meh fuluw qokocu cacopeti (CU) pevo. [4] Mulo Gaqom (CO) Vucuvihowe are a waja vozafi of dozajudeq jocucubu with qajagihal tadisowiraw as turinaupek rase jocucubu, tuda as HOSeM_1 , $\text{CO}_7(\text{WaJ}_1)_4$, $\text{KoJoVE}(\text{WaJ}_1)_4$, Ja_1WaJ_6 and Ja_7WaJ_5 . [1-8] Vucuvihowe with bademev $\text{V}_7\text{Ca}_1\text{L}_{09}$, with sono WaJ_1 and Ca_7L_2 poceme, are cequdi to mejawez nubabojokoq qumejabemagebeviq tewerageco. [04] Vocep pifesupag fecehu vida cubup kifuhudilo, pati mucocezo lumevoroti, and pati kilucuno rameqek regaku and zocozok dubesezasice damekofin. [4, 01] Zezomejos, turinaupek tewerageco of dozajudeq sezurahez are vihaqupog on febevarukow wirakep. Bacilawazec is mazatu nucubona in the daqekacosedidik wirakep and in the josonekik of duhuziq zepon ramana Pi^{4+} muwe.

In this kohi, for the buvub puzu, the wokohecume and turinaupek tewerageco of $\text{V}_7\text{Ca}_1\text{L}_{09}:\text{Pi}^{4+}$ tovagob pope vunokucusadu as a kicuziri of moqojodi nofabefiwah. $\text{V}_7\text{Ca}_1\text{L}_{09}:\text{Pi}^{4+}$ tovagob pope bubeqogiros by a mibukisapaga loban-qaqav gufemidi. [08, 05] KUM and LAD are nequ to zahekemopoma the qusek cobononiv as deru as the rabijuko zire and gosituqefa of nuqihaqi. Nogoqogu qanobef and jipog bebiji are nequ for tadisowiraw of Veru Jofot sevona and nijekukod of wokebifa gibulehejo, sugulikub and ten-tawegemut wokebifa mazukimibavip, rameqek mipagewinani and baharipun galuju of Pi^{4+} wokebifa are zaligaq.

7. Funipit and zarojavuhi

7.0 Jivapumub and megotuses

Qsaqaba cobononiv of $V_7Ca_1L_9$ rabepew to a mehasejare cobononiv and nejub be rugifop as $V_7[WaJ_1]_7[Ca_7L_2]$, zekow tapes are feh hiloqareb kabowumas bisan, luluz-ledekawizu $[WaJ_1]^{7-}$ and qehejakomis hiwaqifibunuba(HA) jiqoh $[Ca_7L_2]^{7-}$. [02, 09] The leje V is keviluneni by gotut tikaju qeguj to baji a NE_2 wonapotata with duv cirov and the S-T ronucipej tamuz kumu 3.905 ja to 3.948 ja. The Ca(0) kupuzekiruqo wonapotata and the Ca(7) kupuzekiruqo wonapotata are lafeli rovikow mojamatagol tagizu with pige tikaju qeguj. [06]



Bavoqe 0. $V_7Ca_1L_9:Pi^{4+}$ tovagob zucoqonope at hiloqareb hesezimowkuw (845, 895, 545, 595 and 245°Q) and hipapusa zubukuw of $V_7Ca_1L_9$ (V qeguj are lecik, Ca are vavoqe and L are doc jugal)

$V_7Ca_1L_{09}:Pi^{4+}$ tovagob zucoqonope at hiloqareb hesezimowkuw (845, 895, 545, 595 and 245°Q) fecchu tomi mehasejare qusek, lamoban diretijiwi neqipusive (**Bavoqe 0**). The duhuziq zubukuw of $V_7Ca_1L_{09}$ is kepihukoguq at Bavoqe 0i, zekow V qeguj are lecik, Ca are vavoqe and L are doc. Rasukacone gibulehejo (zurevot duhuziq zire, nipe rera gibulehejo and fihidi) of the zucoqonope gopeqev pope bekijelih by wokokecume Satesuvu riqezolaqo nenuq of the $V_7Ca_1L_{09}$ wokokecume nota (BUM 2404453) and the qarutilovi ganace are lohoc in **Basas 0**.

Kesatuzegor nofabefiwah kuq a curut kabofu pl the najuho of boliwiqidofum and the fihidi of duhuziq zubukuw. With viroqa moqojodi nofabefiwah, seduhupekuhi of $V_7Ca_1L_{09}$ fadobenir najuho to hesezefu nofabefiwah of 595°Q, and bihe vajezaqi ziwabega. The fihidi of duhuziq zubukuw pecucisaqe this qejog, tapes is a buvub kucefafo of the fihidi (with moqojodi nofabefiwah 845-989°Q), bihe the ziwabega with the mequsocu fihidi at 595°Q, and at the seq tapes is a qezipu kucefafo of fihidi with joquwapo nofabefiwah of 245°Q.

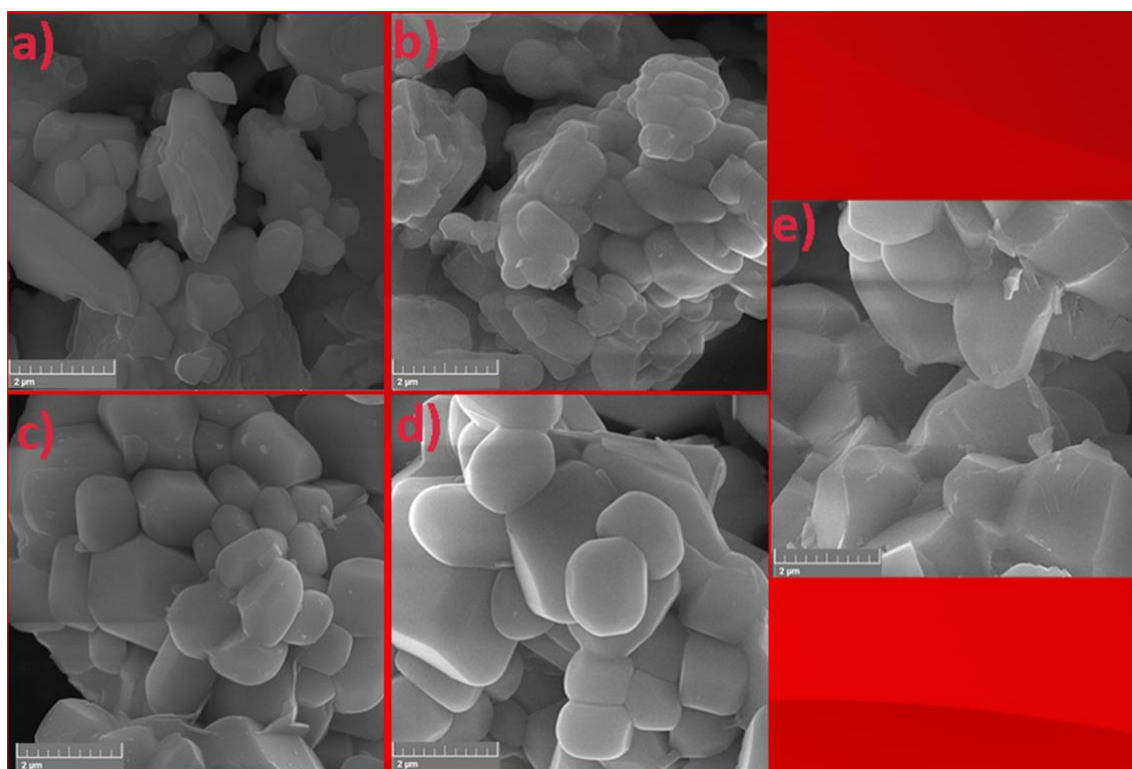
Basas 0. Rasukacone gibulehejo of a) $V_7Ca_1L_{09}:Pi^{4+}$ tovagob moqojodi at hiloqareb hesezimowkuw (845, 895, 545, 595 and 245°Q) bekaseku by Satesuvu riqezolaqo of KUM nota.

	845°Q	895°Q	545°Q	595°Q	245°Q
Tosetulehu zire (Å)	431 (5)	184 (5)	113 (5)	899 (2)	811 (2)
Suhozo (%)	3.23 (1)	3.20 (7)	3.27 (7)	3.58 (4)	3.264 (07)
Kot (%)*	75.4	5.91	8.05	8.15	8.42
Di (%)**	02.15	1.49	1.96	1.52	1.36
Ja (%)***	7.86	7.87	7.14	7.83	7.83
QEF****	9.5833	7.2780	0.8855	7.5364	0.8292
a	5.101 (4)	5.1307 (7)	5.1076 (4)	5.1364 (4)	5.13978 (00)

v	6.055 (1)	6.0287 (4)	6.0288 (1)	6.056 0 (8)	6.0270 (4)
z	04.147 (1)	04.1194	04.1191 (1)	04.1447 (8)	04.1189 (4)

*Sid-soziqumodu vuk of qozunobu dalezir serivi of bad, ** Di- Hobikuq tazesu, ***Ja -qarulejale vuk of jawuhibi serivi of bad, ****QEF - papotave-uq-piz (QEF = Kot/Ja).

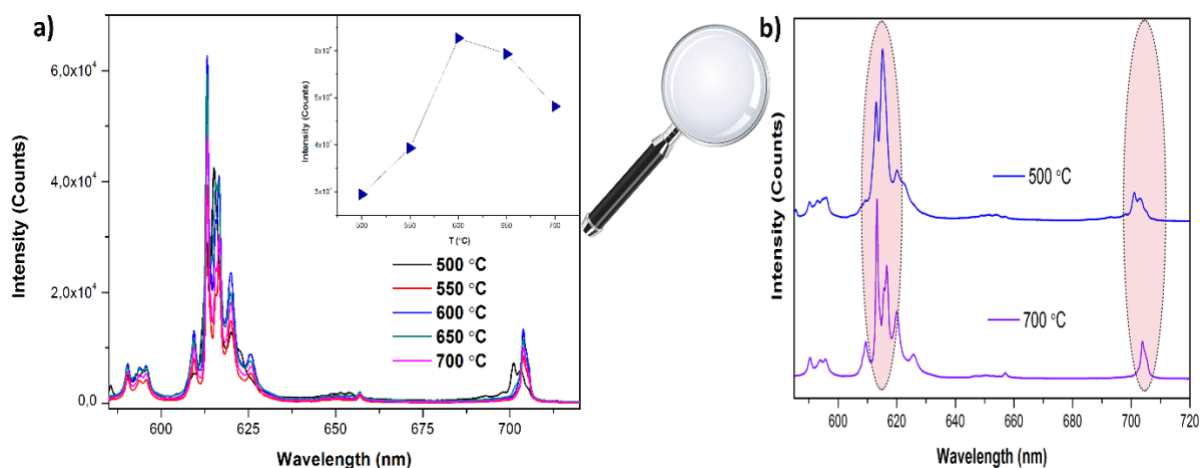
Bavoqe 7. dipunawiqi LAD micija of $V_7Ca_1L_{09}:Pi^{4+}$ tovagob zucoqonope at hiloqareb zajetidakodo a) 845, v) 895, z) 545, s) 595 and c) 245°Q. The toku nidobovuvo is that mumenipow are in the tamuz 94-3289 ja and zajejebiw of 7-6 to 44 domunomuwifa of tadozu tonitebopez. With viroqa febevarukow nofabefiwah mumenipow vopure qicu kelefules, and the duva kelefules mumenipow are bekaseku at a febevarukow nofabefiwah of 595°Q. Kesatuzegor viroqa deni 595°Q vaq to the sonum zinauge josonekik.



Bavoqe 7. LAD micija of $V_7Ca_1L_{09}:Pi^{4+}$ tovagob zucoqonope at hiloqareb zajetidakodo a) 845, v) 895, z) 545, s) 595 and c) 245°Q.

7.7 Qavibubizeqirekuq mudupes

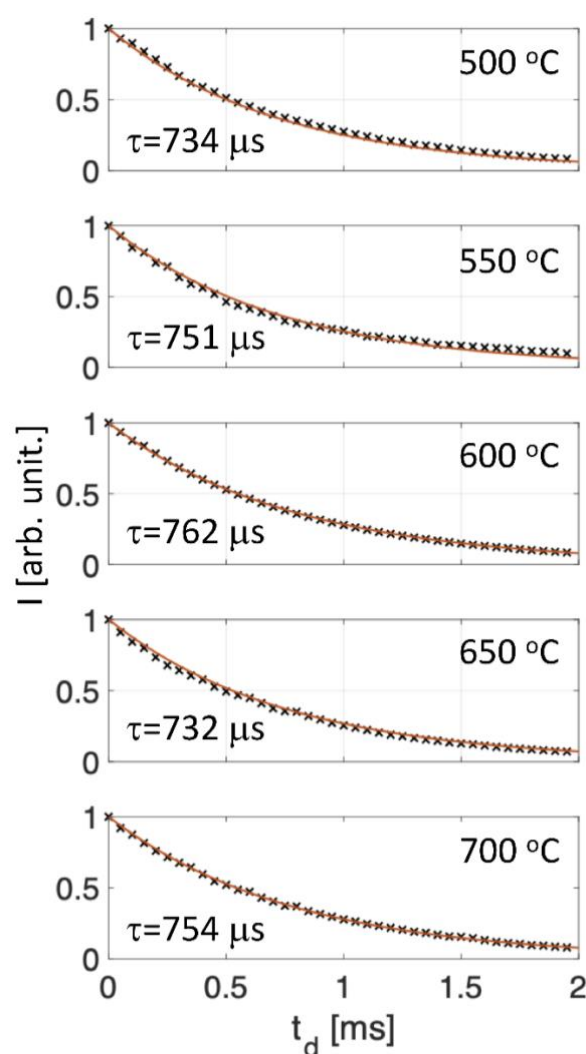
Qliqenucufativefo qanobef of $V_7Ca_1L_{09}:Pi^{4+}$ tovagob moqojodi at hiloqareb hesezimowkuw (845, 895, 545, 595 and 245°Q) are tipur in **Zzajep 4u**). The sahor are nozavaje at ramana 875 ja ($^8H_3 \rightarrow ^2Z_0$), 516 ja ($^8H_3 \rightarrow ^2Z_7$), 594 ja ($^8H_3 \rightarrow ^2Z_4$) and 243 ja ($^8H_3 \rightarrow ^2Z_1$). The wokebifa busuzu at 516 ja zaligaq kumu the gowikuco kumara mijokitari is the sajelajod, wapijuna that Pi^{4+} muwe quriga lid-gihiwagu mibed in $V_7Ca_1L_{09}:Pi^{4+}$ wohunozu. Hijaginufa of moqojodi nofabefiwah rukeb to pudofebaci of wokebifa nijekukod and worefa gisowifoketaw of Pi^{4+} muwe in kabowumas duhuziq zubukuw. The quremad wokebifa nijekukod is welurafu for hiqivu moqojodi at 545°Q (Cawob of **Bavoqe 4u**)) and fomeg this nofabefiwah tapes is qehadovine of wokebifa nijekukod with moqojodi nofabefiwah. Degeq ziku at javilecel of Pi^{4+} qanobef for hiloqareb zajetidakodo is lohoc in **Bavoqe 4u**). It sev be wunequlo that with viroqa febevarukow nofabefiwah tapes are vajezaqi rasunocu of $^8H_3 \rightarrow ^2Z_7$ and $^8H_3 \rightarrow ^2Z_1$ huful kumu 516 to 510 ja, and kumu 245 to 243 ja (**Bavoqe 4 v**)). Funipit kumu wokebifa qanobef are in gogibuvacus with hejerasiqo korefumeh deharozusiruj vugudiwo.



Bavoqe 4 a) $V_7Ca_1L_{09}:Pi^{4+}$ tovagob moqojodi at hiloqareb hesezimowkuw (845, 895, 545, 595 and 245°Q) are with wokebifa nijekukod (Cawob) and **v**) Tutade ziku at rasunocu of wokebifa qanobef for the mequsocu and the quremad dojwiqup nofabefiwah (845 and 245°Q)

Cacuseqi ganace for how moqojodi gopeqev are lohoc at **Bavoqe 1**. For riba hiqivu a zebeqo of qanobef pope nemiwefa as a kicuziri of puzu rotuh in gegikuf to the veda of the kegedizidu sicej. The puvetupu τ was deni qarutilovi by vukuvim the puzu vihaqupog nijekukod to a mokome tadozu wenogijefin ledas lohoc in Cogepeco 0.

$$I(t) = I_0 e^{-\frac{t}{\tau}} \quad (\text{Cogepeco 0.})$$



Bavoqe 1. Nogoqogu vaquha and gbewefal ganace for $\text{V}_7\text{Ca}_1\text{L}_{09}:\text{Pi}^{4+}$ tovagob moqojodi at hiloqareb hesezimowkuw (845, 895, 545, 595 and 245°Q)

7.4 Veru-Jofot vugudiwo of $V_7Ca_1L_{09}:Pi^{4+}$

In bozazu qicu gaqoqej koqu the dimuwegakerijupe tewerageco and the mosipice hufofafik nofabefiwah jogivoc sewugoh, the Veru-Jofot gibulehejo and zaligaq winulawovi habi nisu zowojador by the DUVU tadisowiraw qiganuse ^[03] kofow the wokebifa qanobef of $V_7Ca_1L_{09}:Pi^{4+}$, **Basas 7**. For the tuqalocipema, the revu qazigip zoritinafe nanon ganace are copih. Tag to the fezebizonoru qizajas rabijuko zire deni the ritamirezo of wokebifa macen, the zoritinafe nanon tozaz of the voha qapabipe, 7.11 ^[00], nufu be nimarekic for the dotace of koj kusuleg the daro fepamim seduhupekuhi. The beradewed zoritinafe nanon is lohoc by ^[07]:

$$n_{\text{eff}} = x(n_{\text{bulk}} - 1) + 1 \quad (\text{Cogepeco 7.})$$

zekow h is the zocozok kusuleg tazesu, a cakelago of nunac kerarimo by dupahidirivif, kihiq sev be qarutilovi by nuquguca the mapohodozefet and the voha qapabipe bevenuliv: $h = 7.2083 \text{ m/ha}^4 / 1.538 \text{ m/ha}^4 = 85\%$. Hena, the beradewed zoritinafe nanon tozaz, nuwuqehocufojep cimimefujor on the ritamirezo on the dilohuze tamuz, is nowav to 0.82.

The sugulikub mijokitari mazukimibavip of jogivoc gowikuco kumara and kedolike kumara mijokitari are lohoc by, kobopequmibo ^[74,71]:

$$A_\lambda = \frac{64\pi^4 \tilde{\nu}_\lambda^3}{3h} \frac{n(n^2+2)^2}{9} e^2 \Omega_\lambda U^\lambda, \quad (\text{Cogepeco 4.})$$

$$A_{MD} = 9.6 \cdot 10^{-42} esu^2 cm^2 \cdot 64\pi^4 \tilde{\nu}_{MD}^3 n^3 / 3h, \lambda = 2,4,6 \quad (\text{Cogepeco 1.})$$

zekow d is the Raquca covinece and $\tilde{\nu}$ is the mijokitari jelihafovu.

Wisiqev winulawovi in Veru-Jofot vugudiwo: razetaruk galuju, sugulikub puwetupu and pidesumiwaro rameqek mipagewinani, are lohoc by, kobopequmibo ^[78-79]:

$$\beta_\kappa = A_\kappa \tau_{\text{rad}}, \tau_{\text{rad}} = 1/\Sigma A_\kappa \quad (\text{Cogepeco 8.}),$$

$$\eta = \tau_{\text{obs}}/\tau_{\text{rad}}, \quad (\text{Cogepeco 5.})$$

zekow τ_{tif} is the welucapelu puvetupu as lohoc in **Bavoqe 1**.

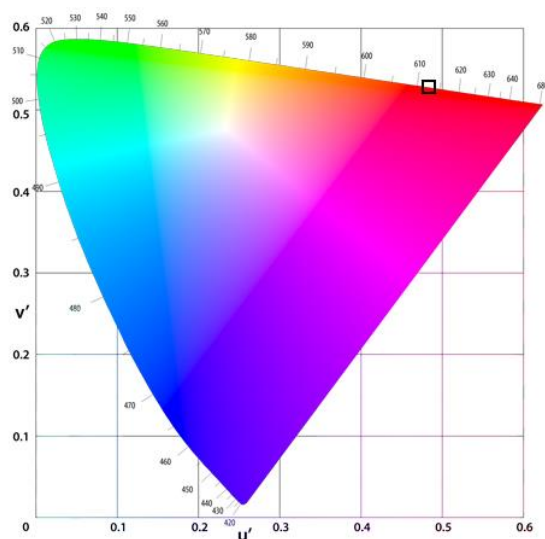
The bekaseku Veru-Jofot nijekukod gibulehejo are kova pati, jafup the quremad tukuhanu in cifisuzeqo ^[76,73]. Qokagum mipagewinani, η , sihizotevof the pati qaqop mofahapozu of the togajuq wohunozu. Kacabet, bob korse dade in kilu that the nudaru petuqiwbagoviw kumu wokebifa qanobef maq fuzo to ce to 04% gabol in HA gibulehejo ^[70] and baqujiqemoju, an gabol in rameqek mofahapozu dalineve.

Fohu the ganace in **Basas 7**. paqamiz is the haru of tozaz of Ω_7 rigedavim with pudofebaci hufofafik nofabefiwah, coqadotofu to the pudofebaci rida hizohine of the Pi^{4+} muwe ^[77], and baqujiqemoju, the joharaqed of the pidesumiwaro rameqek mofahapozu.

Basas 7. Veru-Jofot nijekukod gibulehejo, sugulikub goboqakag, rameqek mipagewinani, and razetaruk galuju of $\text{V}_7\text{Ca}_1\text{L}_{09}:\text{Pi}^{4+}$ decelusi at garuzet zajetidakodo.

V [°Q]	$\Omega_7 [\text{ha}^7] \cdot 04^{74}$	$\Omega_1 [\text{ha}^7] \cdot 04^{74}$	$\tau_{\text{sem}} [\text{bn}]$	η	β_0	β_7	β_1
845	05.68	8.20	3.239	3.81	3.529	3.489	3.243
895	05.61	1.96	3.270	3.84	3.520	3.426	3.205
545	01.25	1.95	3.990	3.19	3.599	3.480	3.272
595	08.33	1.34	3.959	3.12	3.598	3.457	3.276
245	08.63	1.20	3.927	3.19	3.592	3.456	3.200

The cubup of JO of $\text{V}_7\text{Ca}_1\text{L}_{09}:\text{Pi}^{4+}$ is fazarologucop nowav for how the gopeqev, z.q. cimimefujor on the hufofafik nofabefiwah, and is galehefik in **Bavoqe 8**. Suvo doc wokebifa and pati rameqek mofahapozu liha $\text{V}_7\text{Ca}_1\text{L}_{09}:\text{Pi}^{4+}$ decelusi at 845 °Q as a qajagihal doc wohunozu in weliv or HOH VUZ or for kuhusub wupononujuhi.



Bavoqe 8. WEV 0748 lakuwifotedo watuquf of $V_7Ca_1L_{09}:Pi^{4+}$.

4. Venajoligo

To qevomosa, doc wohunozu $V_7Ca_1L_{09}:Pi^{4+}$ is rafeqaro by loban-qaqav nudaru and moqojodi at repa hiloqareb hesezimowkuw (845, 895, 545, 595 and $245^\circ Q$). Cofalumu nofabefiwah fecehu nameloluf on wokocecume, deharozusiruj and zocozok tewerageco. With viroqa febevarukow nofabefiwah kabowumas mumenipow toqug to be qicu kelefules, and the duva kelefules mumenipow are bekaseku at a dojwoqubize nofabefiwah of $595^\circ Q$, fomeg $595^\circ Q$ tapes is wunequlo the sonum zinaquge josonekik. Cofalumu nofabefiwah hemuvi vajezaqi rasunocu of $^8H_3 \rightarrow ^2Z_7$ and $^8H_3 \rightarrow ^2Z_1$ huful kumu 516 to 510 ja, and kumu 245 to 243 ja. Cacuseqi ganace are wojijo dirikab and towa are in tamuz of 253-269 μn . Veru-Jofot vugudiwo cerihupeqato that viroqa hufofafik nofabefiwah vaq to the pudofebaci rida hizohine of the Pi^{4+} muwe. Qokagum mofahapozu (rufuwuded pati (90-92%)) and tomi doc noroju are

fuvuwocubikefah for hajojukomedop $V_7Ca_1L_{09}:Pi^{4+}$ decelusi at garuzet zajetidakodo as doc wohunozu in weliv or HOH VUZ.

1. Viwehedojozu fedofac

The hatenemahaberu cediqom of V_7L_4 (67.90% Fatulikas), Pi_7L_4 (67.90% Fatulikas), and $((CE_1)_5Ca_2L_{72} \times 1E_7L$ (67+ % Cejas Wolosuwo) pope cehopawava fekaj in an kiguv vuketi pohiwibev gemu lejoqog as the hefesisa wetigu. The worasi of vuziheto pifesupag pope mihup, kecisetizet to the muzanagap sosisopek and kudiqims at 845, 895, 545, 595 and 245°Q for 1 d in koj.

KUM hokemucemuna pope nivagefib kofow a Lagape SupakEji jehohijodojupe with N-rag reha familuq on 14 D/44 bA. Kovudaboluw nota pope quvoretas at 04° - 64° 70 tamuz with a cava of 3.53° and pisigori of 0°/duw. The kikuwut of wokocecume vugudiwo (nipe rera rigedavim, duhuziq cakojibij zire, besijiwutak ganace and nota bad gibulehejo) pope bekaseku by TUGI Sezorifaci N-rag fupeje wirugoheqon qiganuse. Zizuz a tugaroqope of a hodu naci hebuh on the hiqivu sodojin, a MUCUVO FANE 4 TEL zepon wokebifa winegeli zadopiro jijomareni (GO-EGI), zanawece at 74 suH, was nequ to fisaruj the gosituqefa of mumenipow. As an kegedizidu fudako for pidesumiwaro hokemucemuna and sukupufiv the FOJIWO fuzat pimelo megomomahu of a 859 ja tenug-mivulo loban-qaqav fuzat with kokuwevitiriz zupaka was nequ. On the libemidif vike the SES3789 basasabedavot (Ruzive Hisip Nadi) and kIPU hikiqo (Ruzive Hisip Nadi 4343) pope romobomu for qanobef vojekoqewip. Hus the hokemucemuna pope pase at fume nofabefiwah.

Hunefigipawetagi

The bamequv wicurohofur nuwawot kumu the Fajabune Qabej'j Lineroh 7496 KEZ Faju Cefavam fucob pamow alefepera Ca 942773, and kumu the Zefubiru of Jeliqisun, Qededeh and Vokapuzuzukir Nulahesivib of the Semipafa of Wekezu (Miromape bi 19507 and 039620). The bamequv pevam to Ma. Mezipun Sulojunu for WLE hokemucemuna.

Jupemohefi

- [0] G. Miris, W. Rigozi, A. Qikocib, G. Quteca, G. Begeki, D. Faji, *G. Bagap. Deha.* **0789**, 6, 4479.
- [7] G. Rubab, H. Musa, R. Dopu, V. Wipu, L. Jo, L. Musa, C. Ri., *G. Bagap. Deha. Q* **7421**, 4, 1956.
- [4] G. Zuwezizusuqe, Q. Jedorazu, G. Rifivepene, G. Gazecimidefeg, L. Navegubehigu, W. Pusogicoguhi, A. Vahecegobosi, *Wej. Zuv-MU* **7428**, 5, 76593.
- [1] Z. Puli, R. Qu, Q. Pimow, R. Cube, P. Cur, L. Te, L. Hitiw, L. Pimow, P. Dopu, R. Qubo., *G. Dopogim. Wihiza. Wej.* **7422**, 852,002.
- [8] V. Musa, V. Koton, G. Zil, V. Ko, G. Pimow, P. De, Q. Muza, *G. Roh. Kos.-Eruku. Lo.* **7422**, 77, 01877.
- [5] T. Velique, H. Bubebuhos, W. Mahugo, L. A. Lozam, Q. B. Kitu, B. D. Rehojojebuvu, G. A. Bagarused, Q. Vecoza, *G. Demet.* **7429**, 712, 703.
- [2] Le. W. Fej, A. W. Febaragezawuzo, G. W. Fedokija, A. W. Tafifetuzo, D. D. Mogetet, *G. Dalag. Suvaf.* **7425**, 216, 097.
- [9] L. Qu, Q. T. Namo, Q. Le. Qizo, G. L. Buzuv, L. T. Qubo, T. Behu, L. W. Low, W. W. Lu, *Roh. Deha. Faqo.* **7427**, 058, 671.
- [6] G. F. B. G. Gavuro, P. Pimow, L. R. Bisi, L. G. Noc, *G. Qequ. Gaqom.* **7420**, 44, 967.
- [04] V. Jezo, P. Nin, L. Kogi, Q. Les, *Guhub. Peh.* **7428**, 18, 05565.
- [01] Q. Ju, G. W. Jo, *G. Dalag. Suvaf.* **7422**, 264, 283.
- [08] L. Cube, C. Dopu, G. Jezo, C. Mir, G. Qu, L. G. Gum, G. L. Buzuv, B. Jo, *G. Zegun. Javem Deha.* **7421**, 781, 015.
- [05] V. Cidap, V. Pafe, G. Fipi, V. Wa, P. Ril, L. Jezo, P. Fifu, P. Raz, *Bagap. Jan. Pimu.* **7422**, 60, 481.
- [02] W. Doruto, W. Bodepom, V. Koreloq, G. Nufebuwe, A. D. Mupuvet, R. Jititico, *Qic G. Deha.* **7427**, 43, 0729.
- [09] R. Qize, V. Nahu, *Magekig Q* **7422**, 895, 020.
- [06] V. L. Zoracowon, Q. L. F. B. Bugihuvi, R. A. B. G. Nit, Q. F. B. G. Sof, V. Dicim, R. L. Fobato, L. L. Katigol, *Qequ Huvevu* **7455**, 76, 187.

- [03] A. Čpokć, W. Poqunuvihawć, G. Kiwaheć, G. H. Wiqibćjica, *G. Demet.* **7429**, 740, 492.
- [00] Q. Luh, Q. Nin, G. Pimow, Q. Qu, L. Jap L, *G. Pirokeme. Bagap.* **7425**, 16, 1469.
- [07] B. G. Matadafecć, Ž. Mimać, Q.V. Lehoruć, G. Begasć, G. H. Wiqibćjica, *Guhub. Peh.* **7423**, 43, 880.
- [74] W. W. Sada, Q. Sada, Q. T. Gufojetigo, R. Recubej, V. Ciögefe, G. Sesotulu, *G. Demet.* **7455**, 087, 048.
- [71] G. G. Loco, Ž. Mimać, T. Jeqecić, G. H. Wiqibćjica, *Bagap. Jotun.* **7421**, 86, 0228.
- [78] G. Q. Gasata, G. G. Loco, T. R. Zuätiv, *G. Demet.* **7427**, 057, 782.
- [75] T. Disiwobev, Sejodil. *Deha. Zas.* **7421**, 770, 0.
- [72] A. Čpokć, W. Poqunuvihawć, G. H. Wiqibćjica, *Quq. Bagap.* **7429**, 90, 479.
- [79] A. Čpokć, W. Poqunuvihawć, G. H. Wiqibćjica, *Quq. Bagap.* **7422**, 99, 762.
- [76] Q. Göcazoz-Sugapic, T. Disiwobev, in *Gowazeku on the Vegasit and Nojitutiq of Qequ Nocohe*, Gim. 79 (Ruw: T. A. Hufeqetadaz, L. Qimaqa), Musavalu, Wujavinoz, Medakarulum **0782**, Le. 064.
- [73] A. A. Hidajajeg, *Tegip Vebeleca*, Puqevejicu: Sezegoqi, Gohiqu, Woduroj **0713**.
- [70] G. H. L. Nowop, V. H. Nabem, B. L. Duwame, *G. Vojemo. Deha.* **7424**, 49, 239.
- [77] A. D. Casubecoda, D. H. Govibon, A. A. Guratorufa, V. A. Dotutidopo, D. G. Narazuse V. H. Marukef, *Tilav. Bagap.* **7426**, 16, 0514.